LEARNER GUIDE

Calculating with bounds

Percentages, Ratio, Growth and Bounds · Lesson 9. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

When a calculation uses rounded values, its answer has bounds too. Each value could be anywhere between its lower and upper bound, so you choose the bound of each value that makes the answer largest (or smallest). For a sum or a product, the largest answer uses both upper bounds. For a difference, take away as little as possible: upper – lower. For a division, the largest answer is the upper bound over the lower bound, because a smaller denominator gives a larger answer.

What you are learning to notice

- Write the bounds of each value first.
- Largest sum or product: upper and upper. Smallest: lower and lower.
- Largest difference: upper – lower. Smallest difference: lower – upper.
- Largest division: upper over lower. Smallest division: lower over upper.
- Check: the rounded-values answer should sit between your two bounds.

Moving through it, one step at a time

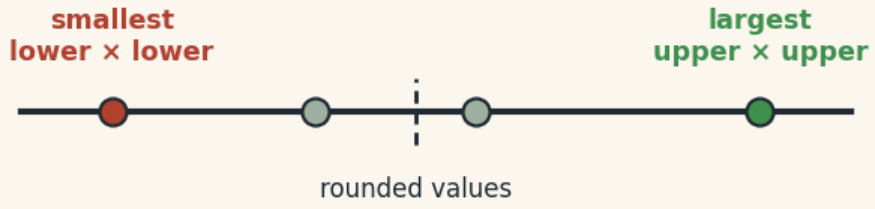
Write each value's lower and upper bound. Decide, for the largest answer, which bound of each value makes it bigger: the bound of anything you add or multiply by should be the upper; the bound of anything you take away or divide by should be the lower. For the smallest answer, swap every choice. Calculate, and give the answer exactly or to a sensible accuracy.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The bounds of an area

A rectangle is 5 cm by 3 cm, both to the nearest cm. Find the upper and lower bounds of its area.

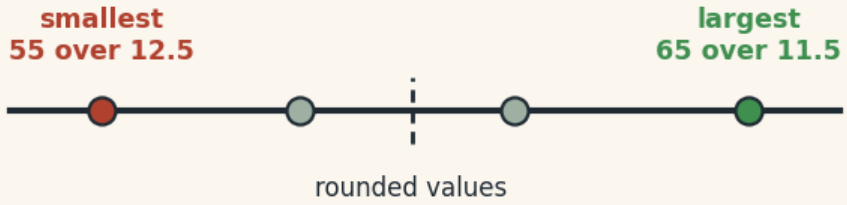


1. The bounds of each side	$4.5 \leq$ length < 5.5; $2.5 \leq$ width < 3.5
2. Largest area: upper × upper	$5.5 \times 3.5 = 19.25$
3. Smallest area: lower × lower	$4.5 \times 2.5 = 11.25$

Upper bound 19.25 cm^2; lower bound 11.25 cm^2. (The rounded values give 15 cm^2, between them.)

Example 2 — The bounds of a speed

A runner covers 60 m (to the nearest 10 m) in 12 s (to the nearest second). Find the upper and lower bounds of the speed.



1. The bounds	$55 \leq$ distance < 65; $11.5 \leq$ time < 12.5
2. Largest speed: upper over lower	$\frac{65}{11.5} = \frac{130}{23} = 5.65$ to 2 d.p.
3. Smallest speed: lower over upper	$\frac{55}{12.5} = 4.4$

Upper bound 5.65 m/s to 2 d.p. (exactly 130 over 23); lower bound 4.4 m/s.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (bounds of a rounded value (Lesson 8) and multiplying and dividing decimals), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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